

Jiayi Yang

jy4656@nyu.edu | [LinkedIn](#) | [Github](#) | [Google Scholar](#)

Education

New York University Shanghai

B.S. in Computer Science; Minors in Mathematics and Business

Sep. 2023 – May 2027 (expected)

Shanghai, China

- Cumulative GPA: 3.95/4.00
- Relevant coursework: Machine Learning (A), Machine Learning with Graphs (A), Recommendation Systems (On-going), Databases (A), Data Structures (A), Probability and Statistics (A), Linear Algebra (A).

New York University

Study Away / Fall 2025, Spring 2026

Sep. 2025 – May 2026

New York City, NY

- Relevant coursework: Natural Language Processing (Grad-level) (A), Time Series Forecasting (Grad-level) (A), Algorithms (A-), Decision Model Analytics (A), Ordinary Differential Equations (A).

Publications

- [1] **Jiayi Yang**, Yifang Chen, Yuanfu Sun, Xinyan Ge, Qiaoyu Tan. **GraphMAS: A Systematic Benchmark of Multi-Agent Coordination for Graph Learning**. *Under review at ICLR 2027*. [Paper]
- [2] **Jiayi Yang***, Yifang Chen*, Yuanfu Sun, Jiajin Liu, Qiaoyu Tan. **One Model, Many Graphs: Learning over Attributed Graphs across Heterogeneous Modalities with Vision-Language Models**. *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026)*. [Paper] [Code]

* Equal contribution.

Research Experience

GraphMAS: Multi-Agent Coordination for Graph Learning

May 2026 – Sep. 2026

Advised by Prof. Qiaoyu Tan

- Proposed GraphMAS, a systematic framework for multi-agent graph learning, studying when specialized LLM agents benefit from complementary structural and semantic graph perspectives and how their coordination should be designed.
- Designed a unified benchmark spanning four coordination paradigms and seven representative methods across node classification and link prediction on seven graphs, showing that gains arise primarily from specialist complementarity and structured reasoning decomposition rather than increased communication alone.
- Developed instance-adaptive routing that achieved the strongest accuracy-efficiency trade-off among training-free methods, and further optimized the coordination policy with PPO over frozen specialists, improving average LP accuracy from **79.6% to 87.3%** and NC accuracy from **68.4% to 72.4%**, with transfer to held-out graphs.

Generalizable Search Agents for General and Deep Search

April 2026 – Present

Co-advised by Prof. Jinyang Li and Prof. Qiaoyu Tan

- Identified a generalization gap between agents optimized for retrieval-based question answering and long-horizon deep search, with deep-search agents underperforming on conventional search benchmarks.
- Explored unified search agent frameworks that jointly learn from general-search and deep-search trajectories with supervised learning and RL-based policy optimization to balance efficient retrieval with long-horizon exploration.
- Built a controlled evaluation framework across PopQA, 2WikiMultiHopQA, BrowseComp, and BrowseComp-ZH to measure cross-regime generalization, compare tool-use and search behaviors, and assess whether unified training can close the performance gap between task-specialized agents.

OMG-VLM: One Model, Many Graphs with Vision-Language Models

Mar. 2025 – May 2026

Advised by Prof. Qiaoyu Tan

- Introduced unified attributed graph learning under heterogeneous modality schemas, enabling a single model to operate across text-attributed, image-attributed, and multimodal graphs.
- Proposed OMG-VLM, a unified generative framework that leverages a pretrained VLM as a shared backbone and incorporates graph neighborhoods through target-aware textual aggregation and graph-aware visual representation learning within the VLM's native embedding space, eliminating the need for separate modality-specific architectures.
- Demonstrated strong node classification and link prediction performance across nine graph benchmarks and multiple GNN-, LLM-, and multimodal-learning baselines, outperforming the strongest baselines by 3.38 points on four in-domain graphs and 15.09 points on five unseen graphs, with gains of up to 20.15 points under cross-graph transfer.

Projects

Robust Language Model Fine-Tuning and Text-to-SQL Generation Apr. 2026 – May 2026

Natural Language Processing Project

- Fine-tuned a T5-small model for Text-to-SQL generation and developed structure-preserving augmentation that jointly modified natural-language inputs and SQL targets, expanding the training set from 4,225 to 35,777 examples.
- Achieved 81.84% Record F1 and 80.32% Record Exact Match, improving held-out Record F1 by 7.54 points through augmentation, model selection, beam-search decoding, and execution-based evaluation.

Graph-Conditioned Image Generation Mar. 2025 – May 2025

Machine Learning with Graphs Project

- Extended the InstructG2I graph-conditioned image generation framework with Graph-QFormer attention masking and CLIP-based neighborhood filtering, reducing FID by 4.2% by suppressing irrelevant graph context.
- Designed a mixture-of-experts visual encoder to replace single-path feature extraction, allowing specialized experts to capture complementary visual patterns and improving the alignment between graph context and visual representations.

Honors and Awards

2026 NYU Shanghai Recognition Award (Top 4%) 2026
Dean’s List for Academic Year 2024, 2025, 2026

Skills

Programming Python, C/C++, SQL, L^AT_EX.
Machine Learning PyTorch, Transformers, scikit-learn, TensorFlow, Git, High Performance Computing.
Languages Chinese (Native), English (TOEFL: 5.5/6.0).